Sharp Optimization-Robustness Envelopes for Classical Synthetic Control

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Abstract

The data-driven Abadie–Diamond–Hainmueller synthetic control is an optimistic bilevel procedure: predictor weights index a set-valued donor-weight problem, and pre-treatment outcome loss selects among its solutions. We define sharp post-treatment effect envelopes over every global or outer-loss-near-global solution admitted by that same procedure. These deterministic envelopes are not confidence intervals.

For rational data, we prove that the donor-weight projection of the complete lower solution graph is a finite union of rational polytopes indexed by coordinate matches and compatible donor-hull face/orthant pairs. Finite convex quadratic and linear programs then give exact global optima, exact-optimum effect endpoints, and separate sign-spanning and attained-zero thresholds. Numerical solvers may discover supports, but independent rational KKT and duality checks accept certificates. Exact synthetic examples show that an envelope can span zero before any zero effect is attainable, which is impossible for a convex donor-weight class under an affine effect. For the declared Basque and California Proposition 99 designs, certified results remain negative through the reported loss gaps. These are procedural checks, not replications of published scalar effects. A supporting extension gives a sufficient exact certificate system for standardized positive predictor-weight floors. The contribution is an SCM-specific finite-graph specialization and certified effect image; reliable point solvers and effect intervals over other convex weight classes are established prior work.

1 Introduction

Synthetic control constructs a counterfactual for one treated unit as a convex combination of untreated donors. In the classical data-driven procedure, donor weights minimize a predictor discrepancy under a diagonal matrix of predictor weights, while the predictor weights are chosen to minimize a pre-treatment outcome loss [5, 2, 3]. This nested construction is widely implemented by `Synth`, but it is not an ordinary convex program in the two sets of weights. Reliable computation has long been a concern [9, 7]; Malo et al. [21] formalize the optimistic bilevel problem, establish general hardness, and give a convergent point-optimization procedure.

Point computation leaves a different question unresolved. Suppose the same declared ADH fitting problem admits several global solutions, or several solutions whose validation loss differs by no more than a substantively small amount. Do they imply the same post-treatment effect? A numerical optimizer returns one selection and cannot answer this question by itself. We therefore hold the estimator, data, donor pool, predictor transformation, and tie convention fixed, and optimize the reported effect over the complete admissible solution set.

For outer-loss tolerance $\varepsilon \geq 0$, write the resulting endpoints as

$$\underline{\tau}(\varepsilon) = \min_{(v,w)\in\mathcal{A}_\varepsilon} \tau(w), \qquad \overline{\tau}(\varepsilon) = \max_{(v,w)\in\mathcal{A}_\varepsilon} \tau(w).$$

The interval between them is an envelope. It need not equal the attained effect set, because the bilevel graph can be disconnected. We accordingly separate the first tolerance at which endpoint signs straddle zero from the first tolerance at which an exactly zero effect is attainable.

This question sits between computation and sensitivity analysis. It is not a sampling interval, prediction interval, placebo distribution, or robustness set for causal-model misspecification. Those remain essential but distinct tasks [17,

10, 16]. Nor is the broad idea of an effect range over alternative weights new. Ferguson and Ross [15] bound effects over convex constrained-error neighborhoods of one-level synthetic-control solutions, and Koo and Guo [20] minimize and maximize over a convex population moment class. Our object differs only after preserving the set-valued, finite-sample ADH bilevel restriction.

The paper makes four contributions.

1. We define exact and near-optimal ADH effect envelopes, distinguish sign spanning from attained zero, and give rational examples where the two thresholds differ.
2. We eliminate continuous predictor weights geometrically. The donor-weight projection of the complete lower solution graph is a finite union of rational polytopes. The representation is constructive for two predictors and supplies exact global optimization and effect certificates.
3. We separate numerical discovery from mathematical acceptance. Floating- point solvers nominate supports; exact rational KKT, LP duality, and Farkas witnesses certify every reported claim. Relaxation sandwiches give rigorous bounds when full graph enumeration is combinatorially unrealistic.
4. We evaluate full public Basque and California designs and implement the two closest convex prior feasible sets on the frozen Basque panel. Both public applications are sign-robust; exact synthetic examples, rather than a selected empirical failure, establish the possible pathologies.

The method complements point solvers. It certifies global objective values and all-solution effect claims when its exact checks close, and otherwise leaves the instance unresolved. It is not a general-purpose replacement for MSCMT or the point algorithm of Malo et al. [21], and it supplies no statistical coverage by itself.

2 Relation to existing work

2.1 Classical SCM and computation

The original synthetic-control construction and California tobacco application are due to Abadie et al. [2], following the Basque application of Abadie and Gardeazabal [5]; the software implementation is documented by Abadie et al. [3]. Abadie [1] provides a broad methodological review. The classical constraints make donor weights interpretable convex combinations, but neither convexity of the fixed-v lower problem nor good pre-treatment fit makes the joint data-driven problem easy.

Becker and Klößner [9] develop reliable inner optimization, identify special cases, and implement MSCMT. They show that existing packages can return suboptimal solutions and give an LP feasibility check for a proposed donor vector. Albalate et al. [7] respond to bilevel instability by changing the estimator and decoupling predictor importance from donor weighting. Malo et al. [21] give the closest formal treatment: they write the optimistic bilevel problem, derive corner candidates and donor-simplex lower bounds, prove existence and hardness, and construct an iterative algorithm converging to an optimal point. Their California application already reports the 1980-cigarette-sales corner solution and loss near 2.74366. We use that design as an external benchmark; we do not claim to rediscover the point optimizer or the package failure.

Our question begins after these contributions: rather than reliably return one point, characterize the entire effect image of the global and objective-gap solution sets. The finite face/orthant representation below also gives a structural description of every lower solution, not only an optimal sequence.

2.2 Alternative weight sets and statistical uncertainty

Nonunique one-level donor weights and sparsity are studied by Chen and Li [11]; penalization and regularization can select a unique or more stable estimator [6, 8, 13]. Those are valuable alternative procedures, but selection removes rather than reports the ambiguity of the original ADH graph.

The closest effect-range objects are convex. Ferguson–Ross constrained error contains the one-level outcome-loss sublevel set after a monotone tolerance mapping. Koo–Guo use an infinity-norm moment class and explicitly report the effect interval over that class. Because an affine effect maps a convex weight class to an interval, sign spanning and attained zero coincide in both cases. They need not coincide on the ADH graph.

Prediction intervals [10], imperfect-fit analysis [16], assignment sensitivity [17], and implementation comparisons [22] address different uncertainty sources. We report the optimization envelope alongside, not instead of, statistical and design-based uncertainty.

2.3 Optimization geometry and claim boundary

Weighted-sum scalarization in convex multiobjective optimization is standard [14]; viewing ADH as a weighted sum of squared predictor residuals is therefore not itself a contribution. General multicriteria quadratic efficient sets can be curved [18]. Classical multiparametric QP instead obtains affine optimizers on polyhedral critical regions under uniqueness or active-nullspace definiteness, with projection machinery for degeneracy [23]. Neither framework directly gives the donor-weight projection studied here: ADH varies the Hessian, permits a set-valued lower solution, and then eliminates predictor weights.

This distinction is substantive rather than terminological. A generic parametric-QP optimal-set map need not be polyhedral [19]. KKT/MPEC and lower-value-function reformulations are standard bilevel tools [12], but they preserve an implicit graph rather than proving the finite donor projection below. The bounded primary-source audit located no direct statement of that complete ADH projection or its certified affine-effect image. We accordingly claim an SCM-specific structural consequence of known normal-cone, scalarization, and bilevel machinery, not a generic parametric-QP theorem or universal priority.

3 The ADH optimization-robustness object

Fix one treated unit, $J \geq 1$ donors, $K \geq 1$ predictors, $T_0 \geq 1$ validation outcomes, and a scalar post-treatment summary. Let $x_1 \in \mathbb{R}^K$ and $X_0 \in \mathbb{R}^{K\times J}$ be treated and donor predictors; let $z_1 \in \mathbb{R}^{T_0}$ and $Z_0 \in \mathbb{R}^{T_0\times J}$ be validation outcomes. Let

$$\Delta_m = \{u \in \mathbb{R}^m : u \geq 0,\ \mathbf{1}'u = 1\}.$$

Predictor weights are $v \in \Delta_K$ and donor weights are $w \in \Delta_J$. For a declared affine post-treatment effect $\tau(w) = a - b'w$, define

$$Q(v, w) = (x_1 - X_0 w)' \operatorname{diag}(v)(x_1 - X_0 w), \tag{1}$$

$$\mathcal{S}(v) = \arg\min_{u\in\Delta_J} Q(v, u), \tag{2}$$

$$L(w) = T_0^{-1}\|z_1 - Z_0 w\|_2^2. \tag{3}$$

The primary graph and optimistic outer optimum are

$$\mathcal{G} = \{(v, w) \in \Delta_K \times \Delta_J : w \in \mathcal{S}(v)\}, \qquad L_\star = \min_{(v,w)\in\mathcal{G}} L(w).$$

For absolute tolerance $\varepsilon \geq 0$,

$$\mathcal{A}_\varepsilon = \{(v, w) \in \mathcal{G} : L(w) \leq L_\star + \varepsilon\}.$$

Compactness and continuity make $\mathcal{G}$ and $\mathcal{A}_\varepsilon$ compact, so all endpoint optima are attained. Relative tolerances $L(w) \leq (1 + \rho)L_\star$ are reported only when $L_\star > 0$; the absolute definition remains meaningful under exact fit.

Definition 3.1 (Two breakdown thresholds)**.** The envelope-spanning and attained-zero thresholds are

$$\varepsilon_{\text{span}} = \inf\{\varepsilon : \underline{\tau}(\varepsilon) \leq 0 \leq \overline{\tau}(\varepsilon)\},$$
$$\varepsilon_0 = \inf\{\varepsilon : \exists (v, w) \in \mathcal{A}_\varepsilon,\ \tau(w) = 0\}.$$

Always $\varepsilon_{\text{span}} \leq \varepsilon_0$, but equality requires additional connectedness.

The graph uses the closed simplex $v_k \geq 0$. This follows the mathematical formulation in Malo et al. [21] and permits coordinate corner solutions. A strictly positive domain, a fixed floor $v_k \geq \delta$, or a deterministic selector within $\mathcal{S}(v)$ defines a different procedure and is analyzed separately when relevant.

4 Finite geometry of the projected lower solution graph

Write the donor predictor columns as $x_j \in \mathbb{Q}^K$, the treated predictor as $t \in \mathbb{Q}^K$, the fitted point $y(w) = \sum_j w_j x_j$, and the donor hull $C = \operatorname{conv}\{x_1, \ldots, x_J\}$. Use the outward normal cone

$$N_C(y) = \{q : q'(z - y) \leq 0 \text{ for all } z \in C\}.$$

For a sign vector $s \in \{-1,+1\}^K$, let $O_s = \{q : s_i q_i \geq 0 \ \forall i\}$. For every nonempty face F of C, including C itself, define

$$N_C(F) = \{q : F \subseteq \arg\max_{z \in C} q'z\}.$$

Thus $N_C(C)$ is the orthogonal complement of the direction space of $\mathrm{aff}(C)$ and can be nontrivial when the donor hull is lower-dimensional.

Theorem 4.1 (Finite rational-polytope donor projection)**.** *For rational predictor data, the projection of the complete lower solution graph onto donor weights is the finite union*

$$\bigcup_{i=1}^{K} B_i^0 \cup \bigcup_{\substack{F \text{ nonempty face of } C, \\ s:\ N_C(F) \cap O_s \neq \{0\}}} B_{F,s},$$

where

$$B_i^0 = \{w \in \Delta_J : y_i(w) = t_i\},$$
$$B_{F,s} = \{w \in \Delta_J : y(w) \in F,\ s_i(t_i - y_i(w)) \geq 0 \ \forall i\}.$$

Every branch is a rational polytope. If the validation data and effect coefficients are also rational, the global validation optimum, exact-optimum effect endpoints, $\varepsilon_{\mathrm{span}}$, and ε_0 reduce to finitely many rational convex QPs and LPs, with an unattainable threshold reported as $+\infty$.

Proof. Let $r = t - y$. Fixed-v first-order optimality for projection onto C is

$$q = \mathrm{diag}(v)r \in N_C(y). \tag{1}$$

If some residual $r_i = 0$, then $w \in B_i^0$; conversely every point of B_i^0 is lower-optimal at $v = e_i$ because its lower loss is zero. If no residual vanishes, $q \neq 0$ because v sums to one. Choose the exposed face $F(q) = \arg\max_{z \in C} q'z$. Optimality gives $y \in F(q)$ and $q \in N_C(F(q))$. The face can equal C when C is lower-dimensional. Setting $s_i = \mathrm{sign}(r_i)$ gives $q \in O_s$, hence $w \in B_{F(q),s}$.

For the reverse inclusion, take $w \in B_{F,s}$ with no zero residual and a nonzero $q \in N_C(F) \cap O_s$. Put

$$\widetilde{v}_i = q_i / r_i \geq 0, \qquad v_i = \widetilde{v}_i / \sum_{\ell} \widetilde{v}_\ell.$$

Then $\mathrm{diag}(v)r$ is a positive rescaling of q. Since $F \subseteq \arg\max_C q'z$ and $y \in F$, it satisfies (1). Zero residuals return to the coordinate argument. Rational polytope faces, normal cones, orthants, and sign inequalities give rational branch descriptions, including duplicate donor columns and nonunique donor representations.

On one branch L is a rational convex quadratic. Fixing the active inequalities of any minimizer leaves a nonempty rational linear KKT system; therefore a rational minimizer exists even for a singular branch QP. Two minimizers on the same branch cannot have distinct fitted validation vectors, because their midpoint would strictly lower squared validation distance. Different branches need not share a fitted vector. Intersecting each globally minimizing branch with its fitted-vector equality makes its optimizer set a rational polytope, so affine-effect extrema are LPs. For breakdown values, minimize L on each branch after adding $\tau \leq 0$, $\tau \geq 0$, or $\tau = 0$. Taking finitely many branch minima proves the claim. □

Allowing the full face is necessary. With donor predictors $(0,0)$ and $(1,1)$, treated predictor $(1,0)$, and $w = (1/2, 1/2)$, both residuals are nonzero, but $v = (1/2, 1/2)$ makes w lower-optimal: the normal $(1/4, -1/4)$ is orthogonal to the donor segment and exposes all of C. An exact adversarial suite also covers duplicate donors, affine dependence, constant coordinates, every treated-point and zero-residual position, simplex-boundary predictor weights, intersecting faces, $K = 1$, one donor, singular validation QPs, and multiple minimizers.

The theorem establishes finiteness, not practical enumeration. A full $K = 13, J = 16$ Basque hull has 136 facets and 58,671 nonempty faces; a naive proper-face/sign pass contains 480,624,640 pairs. This is consistent with the general hardness result of Malo et al. [21].

4.1 Tractable special cases

Proposition 4.2 (Two donors)**.** *With two donors, write $w = (t, 1-t)$. For predictor row k, let $d_k = X_{0,k1} - X_{0,k2}$ and $c_k = x_{1k} - X_{0,k2}$. If some $d_k = 0$, every $t \in [0,1]$ is lower-attainable. Otherwise the union of lower solutions is*

$$D = \left[\operatorname{clamp}\left(\min_k c_k/d_k, 0, 1 \right), \operatorname{clamp}\left(\max_k c_k/d_k, 0, 1 \right) \right].$$

The exact near-optimal effect set is an interval with rational or quadratic- surd endpoints, and $\varepsilon_{\text{span}} = \varepsilon_0$.

The proof writes the unconstrained lower minimizer as a convex combination of c_k/d_k with weights proportional to $v_k d_k^2$. Validation loss is a univariate convex quadratic on D and the effect is affine.

For $K = 2$ and arbitrary J with a two-dimensional predictor hull, exact edge KKT conditions partition each rational hull edge at finitely many rational roots. Lifting each attainable fitted segment back to donor weights gives a finite rational-polytope graph including duplicate and interior donors. At a rational positive gap, effect-cutoff QPs and monotone rational bisection enclose both sharp endpoints to any requested width. No floating cutoff comparison is used.

4.2 Disconnected effect geometry

The finite union need not be connected. A three-donor, two-predictor rational instance has exactly two global optimistic solutions with validation loss $41/18$ and effects $-1/3$ and $+1/3$. A rational perturbation makes the negative solution uniquely global. The positive solution enters at absolute gap $1/1000$, so the envelope spans zero after only 0.0438 percent of optimal loss. No zero effect is then attainable; the first one appears at gap $26081/81000$. This exact example separates the ADH graph from every convex weight class under an affine effect.

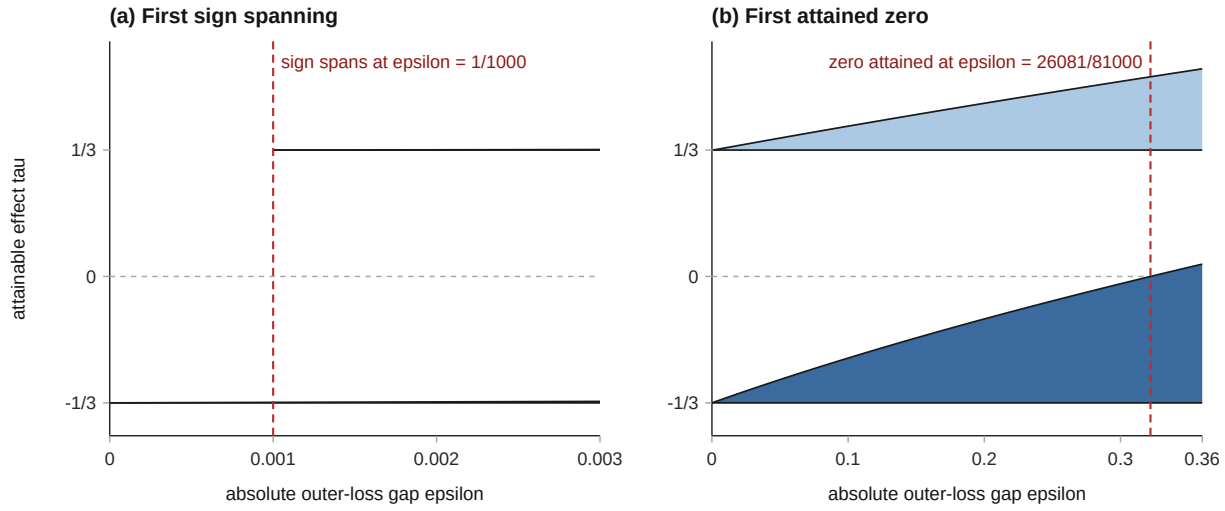


Figure 1: Attainable effects in the exact rational disconnected fixture. Each shaded component is evaluated from its exact branch quadratic, not interpolated between feasible weights. Panel (a) resolves the positive component entering at $\varepsilon_{\text{span}} = 1/1000$ while the negative component remains separated from zero. Panel (b) shows the negative component first reaching zero at $\varepsilon_0 = 26081/81000$.

5 Certificates and scalable bounds

Every constructed branch can be written as

$$P = \{u \geq 0 : Au = b\},$$

where the first J coordinates are donor weights and auxiliary coordinates encode a fitted segment or inequality slack. Validation optimization is a convex least-squares QP and effect optimization is an LP.

Proposition 5.1 (Exact support acceptance)**.** *Suppose a numerical solver proposes positive support S for a rational branch QP. If rational reconstruction yields a primal-feasible u, equality multipliers λ, and reduced gradients satisfying*

$$Hu - g + A'\lambda = 0 \text{ on } S, \qquad Hu - g + A'\lambda \geq 0 \text{ off } S,$$

then u is a global branch optimum. The analogous primal-dual equalities and inequalities certify an LP optimum. Failure to reconstruct such a certificate does not return an approximate claim.

This separates discovery from acceptance. The implementation uses HiGHS and SciPy only to nominate sparse supports. Standard-library rational arithmetic solves the support KKT equations, checks every equality and inequality, and compares branch objectives exactly. Deterministic small cases are also solved by exhaustive rational support enumeration.

5.1 Donor-simplex and coordinate sandwiches

The donor simplex contains the projected bilevel graph. Its unconstrained validation optimum is therefore a global lower bound, and its effect-cutoff optima are lower bounds for all ADH cutoff problems. Any donor vector with an exact lower-level KKT witness supplies the opposite feasible bound. Matching bounds close a global claim, as in the full Basque design.

Even without a general inner witness, every matchable coordinate supplies an exact subgraph B_i^0 at $v = e_i$. Let L_R be the donor-simplex optimum and L_C the least validation loss among coordinate branches. Then

$$L_R \leq L_\star \leq L_C. \tag{2}$$

For relative tolerance ρ, a donor-simplex envelope at threshold $(1+\rho)L_C$ contains the full ADH near set, while a coordinate envelope at threshold $(1+\rho)L_R$ is contained in it. These two calculations give outward endpoint boxes without rounding L_R and L_C into equality.

5.2 Positive predictor-weight floors

The closed simplex admits exact coordinate matches at $v = e_k$. Requiring positive weights changes the procedure, and the open simplex may have an infimum without an optimizer. A fixed standardized floor is closed but depends on the declared standardization. Fix positive rational row scales σ_k^2 and write

$$D(u) = \operatorname{diag}(u_k/\sigma_k^2), \qquad u = \delta\mathbf{1} + (1 - K\delta)z, \quad z \in \Delta_K, \quad 0 < K\delta < 1.$$

Proposition 5.2 (Sufficient exact certificate under a predictor floor)**.** *Suppose the treated predictor is outside the donor hull. For each u in the floor domain, let $y(u)$ be the unique fitted predictor point and let $F(u) = \arg\max_{x \in C}\{D(u)(t - y(u))\}'x$. Suppose a declared finite face family contains every $F(u)$, including the full hull if needed. Require each covered face to be a* donor-labeled simplex*: every donor column on it is a distinct vertex and those columns are affinely independent. Enumerate every nonempty subset S of its donor columns. For every S, construct the exact Cramer polynomials for all support weights, all inactive-gradient inequalities, and the outer-loss difference. If a finite rational simplex partition proves the outer-loss polynomial nonnegative wherever every KKT polynomial is nonnegative, or proves that this KKT-feasible set is empty, then c is a global lower bound on floor-constrained outer loss. An exact fixed-u lower-level witness attaining c proves global optimality.*

Positive $D(u)$ makes the fitted predictor point unique; donor-labeled simpliciality makes each positive donor support rationally parameterizable. Cramer's rule then converts lower optimality and outer loss into polynomial sign claims on the predictor-weight simplex. Appendix B gives the reduction, exact coefficient rules, and adversarial exclusions. The result is a sufficient certificate system: certificate verification is exact, but repeated subdivision is not asserted to terminate or find a certificate.

6 Empirical designs

All exact empirical claims are conditional on documented decimal exports. Source floating precision, data construction, and causal validity are not absorbed into mathematical ε. Table 1 summarizes the two retained public full designs. Effects are treated-minus-synthetic post-period means in the units of each outcome.

Table 1: Certified full-design results

Design	K	J	Global loss	Global effect	5% endpoint conclusion
Basque	13	16	0.004126374	-0.2682005	Both endpoint boxes negative
California 99	7	38	$[2.7436616, 2.7440894]$	$[-19.5710, -19.4563]$	Both endpoint boxes negative

6.1 Full Basque predictor design

The frozen design follows all 13 predictors and 16 controls in the official Synth Basque example, validates on 1960–1969 GDP per capita, and reports the mean 1970–1975 treated-minus-synthetic effect. Direct face enumeration is infeasible. The donor-simplex optimizer passes an exact inner- attainability certificate, closing global loss and effect. The optimum effect is -0.2682004675. Sign spanning and attained zero first occur at a 109.3716 percent relative loss increase. Certified 1 and 5 percent endpoint boxes are fully negative.

The conventional local Synth run has frozen-fixture loss 0.0088647263, 114.83 percent above global, and effect $+0.0003084$. Earlier work already documents implementation failures in this application [9]; the new result is the certified complete-graph effect and robustness certificate, not the observation that local software can be suboptimal.

6.2 California Proposition 99 benchmark

The canonical CRAN tidysynth design uses seven predictors, 38 control states, and 19 validation years. Malo et al. [21] already show on source- precision data that the global point is a corner placing all predictor weight on 1980 cigarette sales, with loss near 2.74366; they also establish the local Synth failure. We freeze a six-decimal rational export and ask the different question of effect robustness.

The donor-simplex/coordinate sandwich gives

$$L_\star \in [2.7436616405, 2.7440893158],$$

a 0.01559 percent relative-width box. Every globally optimal effect lies in $[-19.570993, -19.456267]$. The certified 1 percent endpoint boxes are $[-19.925121, -19.724084]$ and $[-19.135304, -19.050582]$; the 5 percent boxes are $[-20.367147, -20.178410]$ and $[-18.710069, -18.484600]$. A nonnegative or exactly zero effect cannot enter before at least a 31,460.8 percent relative loss increase. The local run remains negative at -18.7197252, but has about 16.9 percent excess loss. California is therefore a robustness benchmark, not evidence that every package failure changes sign.

6.3 Compact screen for a third public design

We also screened a declared standard-fitting variant of the public German reunification panel [4]. It remained strongly sign-robust and added no distinct phenomenon, so Appendix C records its specification and certificate status without promoting it to a third headline application.

7 Comparison with the closest convex feasible sets

The comparison is at the optimization-object level, without Ferguson–Ross placebo calibration or Koo–Guo perturbation inference. For a one-level simplex pre-outcome loss with optimum $L_0 > 0$, Ferguson–Ross constrained error B is equivalent to squared-loss slack

$$\varepsilon = \big((1+B)^2 - 1\big)L_0.$$

Thus fixed-v one-level objective-gap envelopes are not novel. Koo–Guo use a sample moment polytope $\|\widehat{\gamma} - \widehat{\Sigma} w\|_\infty \le q$, which is also convex but uses a different metric.

On frozen Basque data, the Ferguson–Ross optimum and matched 1/5 percent endpoints agree with the ADH certificate to reported precision. At a 1 percent relative squared-loss gap its interval is $[-0.2948300, -0.2395807]$; at 5 percent it is $[-0.3274251, -0.1873163]$. This numerical agreement is not set equality: the exact disconnected example already proves that a convex one-level class cannot reproduce the topology of the ADH effect image in general.

The Koo–Guo class uses a third scale. Its exact sample-moment feasibility- floor effect on Basque is -0.282919. Following the authors' default-style radius construction, the class first spans zero at nominal $\lambda = 0.021$ on their 0.001

Table 2: Feasible-set distinction

Object	Weight set	Tolerance	Zero implication
Ferguson–Ross	Convex one-level outcome-fit sublevel	Relative Euclidean error	Sign spanning attains zero
Koo–Guo	Convex simplex moment-discrepancy class	Infinity-norm moment radius	Sign spanning attains zero
This paper	Generally nonconvex ADH bilevel graph	Outer validation-loss gap	Spanning may precede attained zero

grid. These results establish neither dominance nor a tolerance conversion; the classes answer different robustness questions.

7.1 Claim boundary

Table 3 states the claim boundary explicitly.

Table 3: Claim boundary relative to the closest results

Result	Relation
Reliable ADH point computation	Established by Becker–Klößner and Malo et al.; not claimed here.
Effects over convex weight classes	Established by Ferguson–Ross and Koo–Guo; not claimed here.
Scalarization, KKT, and bilevel reformulation	Established optimization machinery used here.
Finite complete ADH donor projection	SCM-specific structural consequence in Theorem 4.1; no direct statement located in the bounded audit.
Certified effect image over all ADH solutions	Contribution here; headline claims have independent exact replay or explicit enclosures.
General efficient algorithm or statistical coverage	Not claimed; enumeration can be exponential and the envelope is conditional and deterministic.

The strongest defensible category is known general machinery applied to ADH with an SCM-specific structural consequence. The result is neither a new point estimate nor a blanket software-performance claim. It is a certifiable post-estimation diagnostic for the original set-valued procedure: after computing or bounding $L_\star$, ask whether every solution consistent with that fit supports the same substantive conclusion.

8 Limitations and implications

First, the method conditions on the specified donor pool, predictors, preprocessing, validation window, post summary, and optimistic tie convention. It cannot repair an invalid research design. Second, no coverage probability is attached to the envelope. Sampling, assignment, and post-treatment shocks require separate inference. Third, the closed predictor simplex allows exact zeros. Becker–Klößner motivate positive lower bounds. Proposition 5.2 gives a generic exact certificate after a declared standardization, but it is conditional on a complete donor-labeled-simplicial face cover and successful finite subdivision. Fixed floors are not scale invariant, and the certificate search need not terminate.

Fourth, exact rational claims for imported data are conditional on decimal exports. Source binary precision and rounding uncertainty are recorded but not conflated with ε. Fifth, finite graph structure is not polynomial tractability. Full enumeration is credible for small geometry; scalable certificates rely on relaxations, sparse supports, or a small number of exact witnesses. A discovery failure is honest but leaves the instance unresolved.

The public-design results reject two simple narratives. Optimization fragility is not automatic: California remains sign-robust at enormous relative loss, and Basque remains robust through 5 percent. Nor is good local fit a globality certificate: both local runs are suboptimal. Only the Basque local run has a sign opposite to the global effect. Reporting the objective value, predictor-weight domain, certificate status, and effect envelope would make classical SCM analyses more transparent.

9 Conclusion

Classical data-driven synthetic control is a set-valued bilevel procedure. Returning one numerical weight vector can hide both multiplicity and near-optimal ambiguity. We define the corresponding sharp effect envelope, prove a finite rational-polytope representation of the donor-weight projection of its complete lower solution graph, and build exact certificates for global optima, endpoints, and two distinct zero thresholds. The theory yields closed-form and constructive algorithms in small effective dimension and rigorous relaxation sandwiches in larger designs.

The contribution is deliberately narrow. Existing work already supplies reliable point solvers, convex sensitivity intervals, and statistical uncertainty quantification. The missing layer is an independently replayable answer to a procedural question: among all solutions that the declared ADH fitting problem itself regards as global or nearly global, how much can the reported effect change? The public applications answer "not at all in sign" at the reported tolerances, while the exact rational examples establish that a small-gap sign reversal and a gap between sign spanning and attained zero are mathematically possible.

10 Reproducibility

Frozen public fixtures, documented data provenance, exact certificates, and independent replay scripts accompany every headline computational claim. Discovery may use SciPy, HiGHS, or local SCM software; acceptance uses separately implemented rational checks. The standard command is

`./scripts/check_all.sh`.

The optional dependency suite reruns scalable searches and public-data reconstruction. A manuscript-consistency check compares reported values with certificate records, while the release audit scans tracked files, generated PDF text and metadata, and the archive manifest for excluded material.

Acknowledgment of generative-AI assistance

Generative AI tools, including OpenAI GPT models and DeepSeek language models, assisted with literature organization, proof exploration, drafting, implementation, and testing. Computational claims are supported by the exact certificates described in Section 10; AI outputs are not mathematical evidence.

A Additional proofs

A.1 Existence and monotonicity

For each v, $\mathcal{S}(v)$ is nonempty and compact by continuity of Q on Δ_J. If $(v_n, w_n) \in \mathcal{G}$ converges to (v, w), then for every $u \in \Delta_J$, $Q(v_n, w_n) \le Q(v_n, u)$; continuity gives $Q(v, w) \le Q(v, u)$. Thus $\mathcal{G}$ is closed in a compact product, hence compact. Every $\mathcal{A}_\varepsilon$ is a compact sublevel intersection. Endpoint optima and breakdown constrained minima are attained whenever their sign sets are nonempty. Nestedness of $\mathcal{A}_\varepsilon$ makes $\underline{\tau}$ nonincreasing and $\overline{\tau}$ nondecreasing.

A.2 Exact breakdown calculation on a branch union

Let D be the donor-weight projection of $\mathcal{G}$ and define

$$\begin{aligned} m_- &= \min\{L(w) : w \in D, \tau(w) \le 0\}, \\ m_+ &= \min\{L(w) : w \in D, \tau(w) \ge 0\}, \\ m_0 &= \min\{L(w) : w \in D, \tau(w) = 0\}. \end{aligned}$$

When the sets are nonempty,

$$\varepsilon_{\text{span}} = \max(m_-, m_+) - L_\star, \qquad \varepsilon_0 = m_0 - L_\star.$$

Each constraint is linear and therefore preserves the rational branch form. Convex QP KKT conditions give rational minima; absent sign sets are reported as unattainable rather than assigned an artificial finite value.

A.3 Coordinate-sandwich endpoint logic

Let $L_R \le L_\star \le L_C$ be (2). The full relative-ρ set has threshold $(1+\rho)L_\star$. Since this is no larger than $(1+\rho)L_C$, the donor-simplex sublevel at the latter threshold is an outer set. Since it is no smaller than $(1+\rho)L_R$, any coordinate-branch point below the latter threshold is a feasible inner point. Therefore the full lower endpoint lies between the relaxation lower endpoint and the coordinate lower endpoint, while the full upper endpoint lies between the coordinate upper endpoint and the relaxation upper endpoint. Rational cutoff bisection turns each endpoint into an outward box.

B Positive-floor certificate derivation

Fix a covered donor support $S = \{j_0, \dots, j_m\}$ and use j_0 as base. Let B contain the predictor differences $x_{j_\ell} - x_{j_0}$ and let $a = t - x_{j_0}$. With $y = x_{j_0} + B\theta$, the equality-constrained inner conditions are

$$A(z)\theta = b(z), \qquad A(z) = B'D(u(z))B, \qquad b(z) = B'D(u(z))a.$$

Positive $D(u)$ and affine independence make $A(z)$ positive definite, so $d_S(z) = \det A(z) > 0$. Cramer's rule gives

$$w_{j_\ell}(z) = \frac{n_{S\ell}(z)}{d_S(z)}, \qquad w_{j_0}(z) = \frac{d_S(z) - \sum_{\ell=1}^m n_{S\ell}(z)}{d_S(z)}.$$

For a singleton support, $m = 0$, the empty determinant is one and the base weight is identically one. The complete KKT polynomial list contains every displayed weight numerator, including the base numerator, and for every inactive donor k the cleared normal-cone inequality

$$-d_S(z)\, q(z)'(x_k - x_{j_0}) \ge 0, \qquad q(z) = D(u(z))(t - y(z)).$$

The sign is fixed by the outward convention $q'(x - y) \le 0$.

For validation residual numerator $E_{S,t}(z)$, exact denominator clearing gives

$$L(z) - c = \frac{P_S(z)}{T_0 d_S(z)^2}, \qquad P_S(z) = \sum_t E_{S,t}(z)^2 - T_0 c\, d_S(z)^2.$$

The denominator is strictly positive. Entries of A and b are homogeneous linear forms after replacing constants by multiples of $\sum_k z_k = 1$. Consequently d_S and the Cramer numerators have degree at most m, inactive gradients have degree at most $m+1$, and P_S has degree at most $2m$; cancellation may lower these degrees. Each polynomial is homogenized to its own actual target degree before coefficient testing.

On any rational simplex cell, nonnegative degree-matched Bernstein coefficients prove $P_S \ge 0$. A KKT cell is infeasible only if every degree-matched coefficient of one required KKT polynomial is strictly negative, with absent monomials treated as zero. The latter rule is essential: $-z_0 z_1$ has one stored negative coefficient but vanishes on the boundary. Longest-edge midpoint subdivision preserves rationality. The verifier rebuilds every cell and accepts only a prefix-free set of leaves that covers the root simplex, including shared boundaries. A missing support or uncovered cell is an incomplete attempt, not a certificate.

The donor-labeled-simplex hypothesis cannot be weakened to geometric simpliciality. A segment containing a duplicate or interior donor column has a nonunique donor fiber even when its fitted predictor point is unique; the outer validation fit may vary across that fiber. Enumerating every nonempty subset handles support boundaries, while exact affine-rank checks reject singular supports.

A rational regression fixture exercises KKT pruning. Donor predictors are $(0,0)$ and $(1,1)$, the treated predictor is $(1,0)$, validation values are 0 for the treated unit and $(0,1)$ for donors, and both declared row scales are $1/3$. Under $u_k \ge 1/4$, the exact optimum is $1/16$ at $u = (1/4, 3/4)$ with donor weights $(3/4, 1/4)$. The first donor alone would have outer loss zero if inner optimality were ignored, but its inactive-gradient polynomial is strictly negative throughout the floor domain. A separate $(z_0 - z_1)^2$ fixture requires one exact midpoint subdivision. These fixtures test the certificate without relying on an empirical application.

C Additional public benchmark screen

The screened German reunification variant uses six published predictor categories, 16 donors, 1960–1989 GDP as validation outcomes, and mean 1991–2003 GDP as the effect summary. This declared standard-fitting variant is

not the article's two-stage cross-validation and refit. An exact donor-simplex/coordinate sandwich places every global effect between −1717.01 and −1256.39 in source GDP units; zero requires at least a 560.3 percent relative loss increase, and donor-simplex outer bounds keep the 1 and 5 percent envelopes negative. The fixture and certificate document the benchmark, but it adds no qualitative behavior beyond the two main applications.

D Certificate architecture and data provenance

The repository contains four distinct layers.

1. *Discovery*: deterministic searches, SciPy support proposals, and local R package runs. These may suggest a support but prove nothing globally.
2. *Acceptance*: rational LP/QP KKT checks, exact hull/face arithmetic, and exact nonnegative-system or Farkas decisions.
3. *Certificates*: JSON or text records containing exact fractions, sparse witnesses, branch decisions, and endpoint boxes.
4. *Independent replay*: verifier scripts rebuild geometry and compare every saved claim. Standard and optional dependency suites are separated.

The principal commands are

`./scripts/check_all.sh` `./scripts/check_optional_scalable.sh`.

At the paper freeze, the standard suite runs 56 tests: 50 pass and six optional SciPy tests skip by design. The optional suite passes six additional tests and replays the Basque, California, closest-prior, and screened-public-design certificates. The positive-floor synthetic certificate is replayed with only standard-library rational arithmetic. When source data and repository-local R packages are available, all committed public fixtures and local Synth outputs reproduce byte-for-byte. A release audit scans tracked paths, public source content, PDF text, and PDF metadata before an allowlisted archive can be built.

Table 4: Frozen empirical provenance

Design	Source status	Exact-data convention	Redistribution
Basque	CRAN Synth 1.1-10 package example	Six-decimal rational export	Package-derived fixtures committed
California	CRAN tidysynth 0.2.1 package example	Six-decimal rational export	Package-derived fixture committed

All computational timing claims are descriptive. Exact status comes from the certificate algebra, not runtime, numerical convergence flags, or printed decimal agreement.

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